

Cybersecurity for Regular Users

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Contents

1. Protection
2. Information
3. Opportunities
4. References and Licenses
5. Take-home message



1)

Protection

1.1 Cybersecurity in everyday life

- The challenge with cybersecurity is to balance security with usability.
- Best practice steps for prevention of cyberattacks might seem tedious or lead in less productivity of the actual work, but when a cyberattack will happen the results might be tremendous and the steps to cure might need even more time.
- Currently, cyberattacks are not a question of if they will happen to you, but when.
- In Greek/Στα ελληνικά: Συστάσεις προς πολίτες από την Εθνική Αρχή Κυβερνοασφάλειας

1.1 Common cybersecurity myths [1]

- Strong passwords are adequate protection
- Most cybersecurity risks are well-known
- All cyberattack vectors are contained
- Some industries are safe from risk
- Cybercriminals don't attack small businesses

1.2 Cyberattack – Case 1

1. You sign-in to an account using the same password as your email account so that you don't have to remember many passwords.
2. This website/application has been compromised and there is an exposure of usernames, emails, passwords, birth dates.
3. You notice unusual behavior in your email account e.g. login from unusual location.

How to prevent that?

1.2.1 Password Hygiene (1/3)

- Use strong passwords with many digits, symbols, numbers.
- Top 200 Most Common Passwords Don't use 123456 like 3,018,050 users did!
- For a hacker it's all about trying different combinations to find out the real password. When using only numbers, the possible values of each digit are 10 (from 0 to 9). When using only lowercase letters of the English alphabet the possible values are 26 (from a to z). When using both lowercase and uppercase the possible values are 52 (from a to z and from A to Z). When using only symbols e.g. \$, & there are 36 possible values.

How Safe Is Your Password?

Time it would take a computer to crack a password with the following parameters

	Lowercase letters only	At least one uppercase letter	At least one uppercase letter +number	At least one uppercase letter +number+symbol
1	Instantly	Instantly	-	-
2	Instantly	Instantly	Instantly	-
3	Instantly	Instantly	Instantly	Instantly
4	Instantly	Instantly	Instantly	Instantly
5	Instantly	Instantly	Instantly	Instantly
6	Instantly	Instantly	Instantly	Instantly
7	Instantly	Instantly	1 min	6 min
8	Instantly	22 min	1 hrs	8 hrs
9	2 min	19 hrs	3 days	3 wks
10	1 hrs	1 mths	7 mths	5 yrs
11	1 day	5 yrs	41 yrs	400 yrs
12	3 wks	300 yrs	2,000 yrs	34,000 yrs

Source: Security.org



statista

This graph [2] is from 2021, meaning that as years pass and computational power increases hackers can find the password in even less time than indicated.

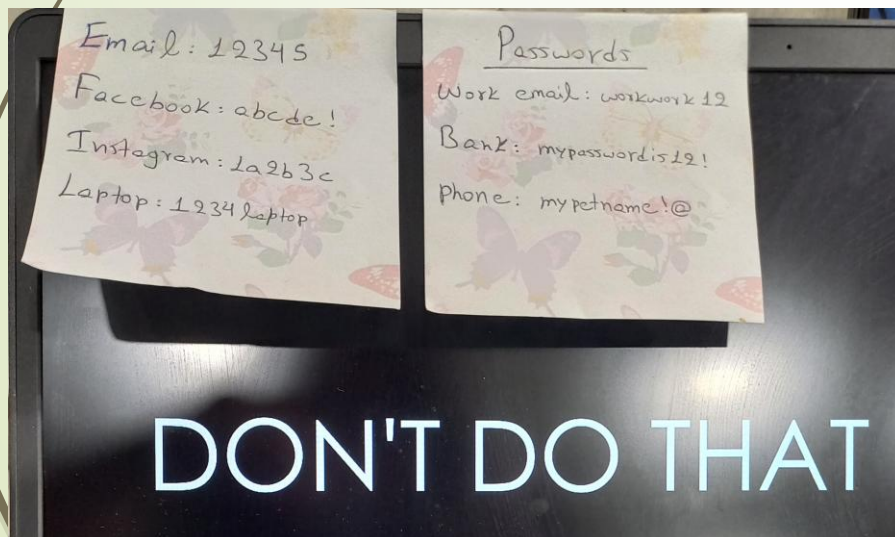
1.2.1 Password Hygiene (2/3)

- Remember that a hacker doesn't manually try all combinations and even a seemingly large number of possible passwords might be easy to crack for a computer. Let alone a computer with much computational power. This is why after some wrong inputs the screen of your phone blocks, so that there is time to wait until next trials and all these combinations cannot be tried fast (brute force attack).

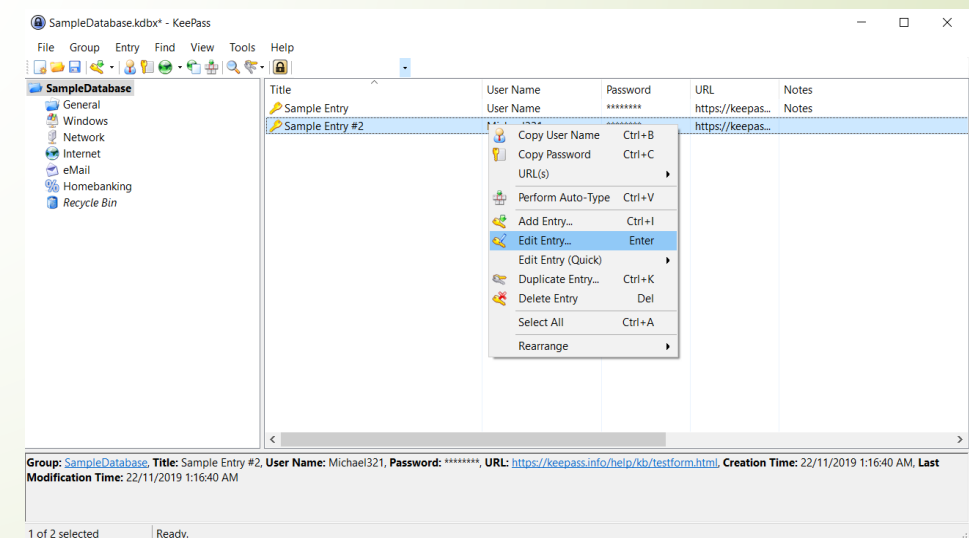
Password Contains	Password length	Possible passwords	Possible passwords
Numbers	4 digits	10^4	10,000
Numbers	8 digits	10^8	100,000,000
English lowercase letters	8 digits	26^8	208,827,064,576
Numbers, lowercase and uppercase letters	8 digits	62^8	218,340,105,584,896
Numbers, lowercase and uppercase letters, symbols	4 digits	98^4	92,236,816
Numbers, lowercase and uppercase letters, symbols	8 digits	98^8	8,507,630,225,817,856

1.2.1 Password Hygiene-Password manager(3/3)

- Do not use the same password everywhere. If there is password leakage only this specific account will be compromised and not all.
- A password manager is a software program to prevent password fatigue by automatically generating, autofilling and storing passwords. It can do this for local applications or web applications such as online shops or social media. Web browsers tend to have a built-in password manager. Password managers typically require from the user to create and remember a single password, to unlock and access the stored passwords. Password managers can integrate multi-factor authentication and passkey authentication. [3] KeePass Password Safe is a free and open-source password manager.
- Google Password Manager application



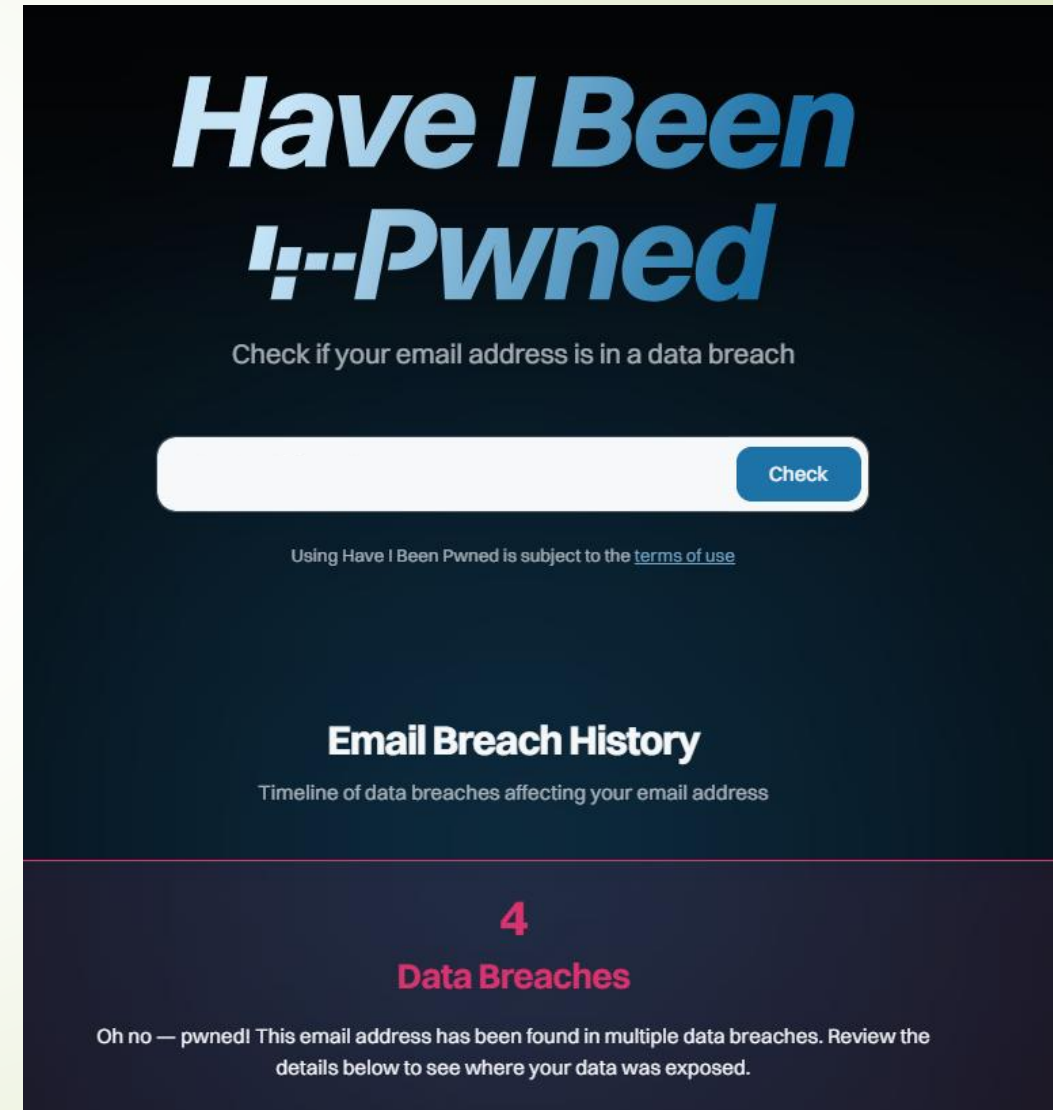
VS



By Dominik Reichl-<https://keepass.info/screenshots.html>, GPL,
<https://commons.wikimedia.org/w/index.php?curid=1635877>

1.2.2 Learn about data breaches

- Check if your email address is in a data breach
- Have I Been Pwned website



1.2.3 Multi factor authentication (1/2)

- Multi factor authentication (MFA) is when another verification method is needed to access an account. E.g. for accessing your email you don't just insert your password but you also have to type the one time password or insert a secure device etc. This decreases the possibility of account compromising even if an attacker has access to your password because they would also need access to other media.
- Security keys are secondary hardware devices that rely on a primary device, such as a workstation, application or laptop. Most security keys today utilize public key cryptography for authentication. During registration, the public key is associated with the user's account. When the user logs in, the service sends a challenge, and the key signs it with its private key, creating a unique signature. [4]
- Google Authenticator is a software-based authenticator by Google. It implements multi-factor authentication services using the time-based one-time password (TOTP) and HMAC-based one-time password (HOTP), for authenticating users of software applications. When logging into a site supporting Authenticator (including Google services) or using Authenticator-supporting third-party applications such as password managers or file hosting services, Authenticator generates a six- to eight-digit one-time password which users must enter in addition to their usual login details. [5]

1.2.3 Multi factor authentication (2/2)

- ▶ There are other ways to login even without your password. Remember the option to reset password when you have forgotten it. What steps should you do? Is a confirmation email sent? A message on your phone number? An attacker could exploit the forgotten password option and change your password if they gain access to the media that is used for this process.
- ▶ Have you enabled the options to connect to an account via other ways except only from your password? E.g. in the email there is the option to message a trustworthy device of yours. Is it a secure device? If an attacker is granted access to this device, which of your accounts could be compromised?
- ▶ Be conscious of your accounts and the ways that someone could login to them.

1.2.4 SIM Swapping

- The attacker typically begins a SIM swapping attack by gathering personal details about the targeted subscriber. There are many ways personal data can be retrieved, this can be done through social engineering, phishing, malware, exploiting information from data breaches or doing research on social media.
- Having all necessary information, the attacker would be able to convince the mobile network operator to transfer the subscriber's mobile number to a new SIM card under their control, or perform the process themselves online. As a result, the attacker takes over the account and can receive all the SMS and voice calls intended for the legitimate subscriber.
- Fraudsters can perform online banking frauds but can also bypass the two-factor authentication (2FA) used to secure social media and other online accounts. [6]

1.3 Cyberattack – Case 2

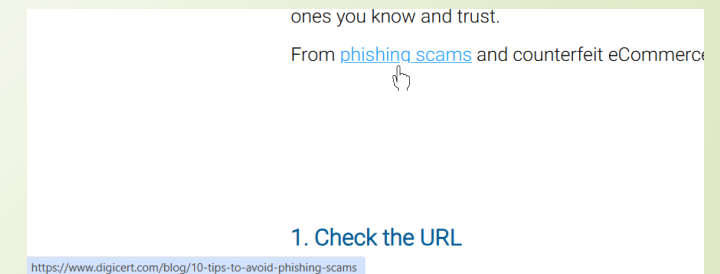
1. You search for information on the internet.
2. An attacker has deployed a fake website and via Spamdexing this website appears to the search results as very relevant.
3. You trust the website because it appeared high on the rank of the search engine and click on different links inside it.
4. This website does malicious things with your information.

How to prevent that?

1.3.1 Spot a fraudulent website [3] (1/4)

➤ Check the URL

- One of the simplest ways to spot a fake website is by carefully examining the URL. Fraudsters often tweak the spelling of familiar domain names in subtle ways, hoping you won't notice. They might swap a letter for a similar-looking character, like replacing the "o" in amazon.com with a zero to create amaz0n.com. Or they could change the domain extension entirely, switching from .com to .org, .net, or another less familiar extension so the URL reads amazon.org or amazon-shop.net. These small changes are easy to miss, especially if you're in a rush. That's why it's important to take a moment to verify the URL before entering any personal information or making a purchase.
- Hover over links in emails or ads to see the full URL before clicking, and always type trusted URLs directly into your browser to avoid being misled by what pops up in search results.



1.3.1 Spot a fraudulent website [7] (2/4)

► Check for a site seal

- A site seal is a visual indicator that a website has undergone verification, signaling to users that the site is authentic and secure. Trusted seals often belong to well-known certificate authorities (CAs) and they are interactive, when you click on the seal you'll see additional details about the website's security measures and how it was verified. This might include the organization's name, the type of security certificate it uses, and the issuing authority. If a site seal doesn't respond when clicked or redirects to an unrelated or suspicious page, consider it a red flag. Fraudsters often copy trusted site seals and paste them onto fake websites to create a false sense of legitimacy. If you encounter a seal that seems questionable, take extra precautions by cross-checking the URL and other security features of the site to ensure the website owner is who they say they are.

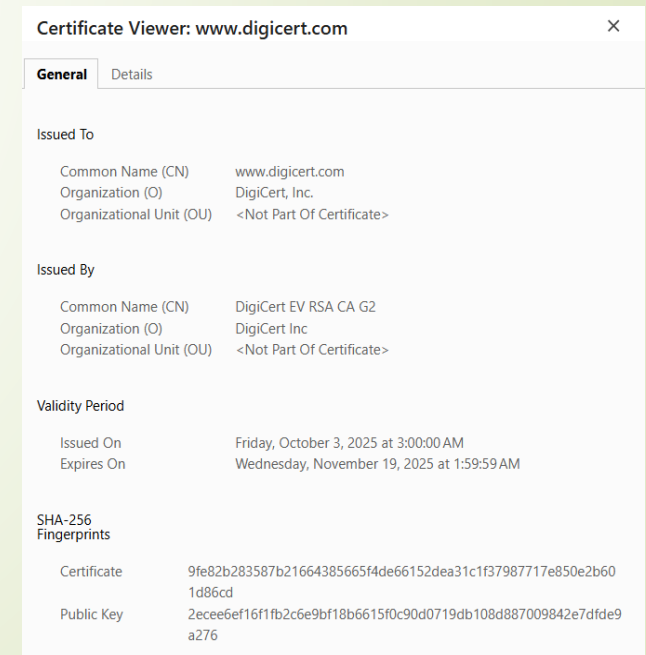
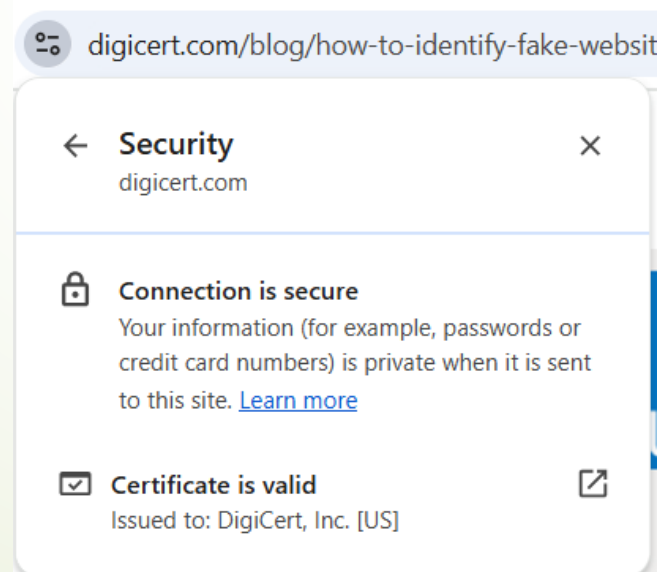
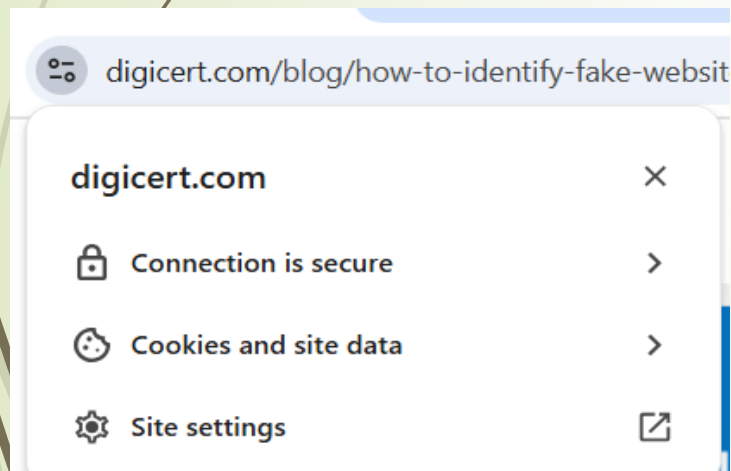
► Use a website checker

- When in doubt, use a website checker to verify if a website is secure. Tools like Google Safe Browsing provide detailed insights into a site's security status. All you have to do is drop the URL into the status checker to see if it's safe or vulnerable to threats like malware, phishing, or weak encryption.
- <https://transparencyreport.google.com/safe-browsing/search> website checker

1.3.1 Spot a fraudulent website [7] (3/4)

► Check the site's security information

- Websites use TLS/SSL certificates to encrypt the data exchanged between your device and the site, to protect it from interception by attackers. When you visit a website, your browser provides visual cues about the site's certificate and connection status. In Chrome (and many other browsers), this is indicated by the small icon to the left of the URL in the address bar, often referred to as the "site information icon," which you can click to get more information about the site's security. While the padlock is a helpful starting point, keep in mind that it doesn't guarantee the legitimacy of the site, only that the connection is encrypted. You can look beyond the lock by clicking on it once, to reveal more information.



1.3.1 Spot a fraudulent website [7] (4/4)

► Look for other trust indicators

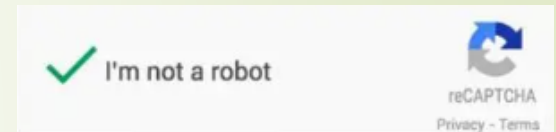
- A website's technical information isn't the only way to tell if it's a safe place to shop. Here are some other markers most legitimate companies have on their websites: privacy policy, return policy, contact information like a business phone number and address, correct spelling and grammar. It's also wise to look at online reviews.

► When in doubt, don't fill it out

- If you discover that a site you've landed on is fraudulent, don't provide sensitive information like financial details, verification codes, a login and password, or your name and contact information. If you land on a fake website, report it to Google Safe Browsing to help keep other shoppers safe.
- <https://safebrowsing.google.com/safebrowsing/report-url> website for reporting

1.3.2 Fake CAPTCHA pages that install malware [8] (1/2)

- A CAPTCHA (Completely Automated Public Turing test to tell Computers and Humans Apart) is a security feature used to distinguish real users from bots. You've likely seen them in the form of distorted text to retype, or image grids where you click on traffic lights or bicycles. The goal of a CAPTCHA is to prevent abuse, such as account spamming, brute-force attacks, or the automated submission of forms.
- Cybercriminals, however, mimic this process by creating fake websites that offer access to movies, music, photos, or news articles, requiring you to "verify" yourself. At first glance, it all looks legitimate: you're asked to tick the "I'm not a robot" box.
- Once you click the checkbox, you're redirected to a set of fake instructions that lead you straight to malware. A typical message might look like this:
 - "To further prove you're not a robot, complete the steps below."
 - Press the Windows key + R.
 - In the verification window, press Ctrl + V.
 - Press Enter to complete the process.
 - You agree to: "I'm not a robot – reCAPTCHA Verification ID: 8253."



Verification Steps

1. Press Windows Button "  " + R
2. Press CTRL + V
3. Press Enter

1.3.2 Fake CAPTCHA pages that install malware [8] (2/2)

- These instructions may seem harmless, but by following them, you unknowingly copy and execute a command from your clipboard in the background. This command often downloads an infostealer or another type of malware. An infostealer is a type of malware specifically designed to steal sensitive data from its victims. This can include stored passwords, browser data, cookies, credit card information, or login credentials for email and social media accounts.
- An increasing number of attacks rely on tricking users into infecting their own devices. Be especially **cautious** when you are asked to use commands like **Ctrl + C** and **Ctrl + V** (copy and paste), or **Windows key + R** (run a command). A legitimate CAPTCHA **will never ask** you to perform these actions.

1.3.3 More on systems protection

- Regularly update software: Outdated systems often contain vulnerabilities that cybercriminals can exploit. Regularly updating applications and patching security flaws ensures they have the latest protections. An endpoint protection agent can monitor operating systems and applications and notify the user of needed updates.
- Firewall
- Antivirus

1.4 Data loss – Case 3

1. You have created a document and you want other people to have access on it.
2. You share the link of the document in cloud and let people edit, review, view it.
3. By mistake someone deletes some information because they don't know that the document is the same for everyone.

How to prevent that?

1.4.1 Privacy Settings

- Who access files and what rights they have. Role based access control
- Limit user privileges: The zero trust model advocates for the principle of least-privileged access. This means users only have access to the resources necessary for their roles, thus minimizing the potential damage if their accounts are compromised.
- GDPR

1.4.2 Back up data [9]

- One of the best backup strategies to have is the 3-2-1 rule. This means creating 3 different copies of your data to put on 2 different storage types and keeping 1 copy offsite.
- USB flash drives are great portable storage solutions to store the most important files from your computer.
- Cloud storage is a great way to create backups as an online media type. This service can store files, photos, or any other type of data. Cloud storage can be used as a primary backup or as a secondary backup. Most providers offer encryption services so that your data is secure while it is stored in the cloud.
- You can back up your data with an online backup service by encrypting your files, scheduling regular backups, and storing the backup files in a safe location. Online backup services are a great way to protect your data in case of a computer crash or theft.

1.4.3 Encryption

- Leverage encryption software: Sensitive data must be protected both at rest and in transit. Encryption helps safeguard confidential information by obfuscating its contents and rendering it unreadable to unauthorized parties, even if it's intercepted or stolen.
- End-to end encryption is used from applications like whatsapp, messenger.

1.5 Cyberattack – Case 4

1. You are using an email account and receive an email tailored to your needs by an unknown but seemingly trustworthy company.
2. You believe that as this company offers you exactly what you were searching before, it's good to examine what they provide and click the link that seemingly leads to their website.
3. Malware is installed on your device.

How to prevent that?

1.5 Phishing and Social Engineering

- Interesting video on how exposed our personal information might be:

<https://www.youtube.com/watch?v=F7pYHN9iC9I>

- **Reporting suspicious emails** can help prevent the impact of a phishing campaign. It's recommended to build a culture that rewards the users who report emails. This gives positive feedback that will encourage users to report more emails in the future.
- **Gophish** is a powerful, open-source phishing framework that makes it easy to test your organization's exposure to phishing.
- Check who sent the email via hovering over the email address before opening the email. Do the same before opening any links. If the URL is suspicious do not click it. Do not open or download attached files.
- **Informing people** on possible cyberattacks and threats is crucial for a good cybersecurity posture. Social engineering targets people that are not aware of the above and provide the attacker with useful information.



2)

Information

2.1 European Cybersecurity Month

- ▶ The European Union Agency for Cybersecurity (ENISA) is partnering with the Commission and Member States in carrying out #CyberSecMonth: the EU's annual campaign dedicated to promoting cybersecurity among EU citizens and organizations and providing up-to-date online security information through awareness raising activities and sharing of good practices.



2.2 Cyber First Aid – Greece [10]

- Are you a victim of an online shopping **fraud**?
- Has your bank account been **compromised** after shopping online? Have you noticed **unusual** payment card activity?
- Has your social media account been **hacked**?
- Contact / Ask for help / Report the incident to:

- <https://cyberalert.gr/>

Δίωξης Ηλεκτρονικού Εγκλήματος, Τηλέφωνο : 11188, Email: ccu@cybercrimeunit.gov.gr, Ταχυδρομική διεύθυνση: Λ. Αλεξάνδρας 173, T.K. 11522, Αθήνα

- <https://www.safeline.gr/>

Safeline, website: www.safeline.gr, Email 1: contact@safeline.gr, Email 2: report@safeline.gr, N. Plastira 100, Vassilika Vouton, 700 13, Heraklion, Crete

- <https://www.gov.gr/ipiresies/polites-kai-kathemerinoteta/kataggelies>

- Get more information from the consumer protection organization in your country:

- <https://kataggelies.mindev.gov.gr/>

- Subscribe here to receive scam alerts / cybersecurity news:

- <https://saferinternet4kids.gr/>

2.3 SaferInternet4Kids [11]

- Talk about the benefits and risks of using AI
- Try AI tools together
- Talk about the potential for misleading or biased content
- Discuss plagiarism and cheating
- Cultivate the child's critical thinking



The poster features the SaferInternet4Kids.gr logo at the top, with the tagline 'ΓΙΑ ΕΝΑ ΑΣΦΑΛΕΣΤΕΡΟ ΔΙΑΔΙΚΤΥΟ'. The main title is 'Artificial Intelligence Tips for Parents' in red and black text. Below the title is an illustration of a red robot with a speech bubble, standing next to a smartphone displaying a chat interface with a robot icon. The background includes faint gear and question mark icons. At the bottom right is the European Union flag with the text 'Co-funded by the European Union'.

help
saferinternet
210 6007686
Help Line


SaferInternet4Kids.gr
ΓΙΑ ΕΝΑ ΑΣΦΑΛΕΣΤΕΡΟ ΔΙΑΔΙΚΤΥΟ
Awareness Center

safeLine.gr
HotLine



3)

Opportunities

3. Competitions and tools

- European Cybersecurity Challenge: <https://ecsc.eu/>
- Capture the flag (<https://www.hackthebox.com/>, <https://ctflearn.com/>)

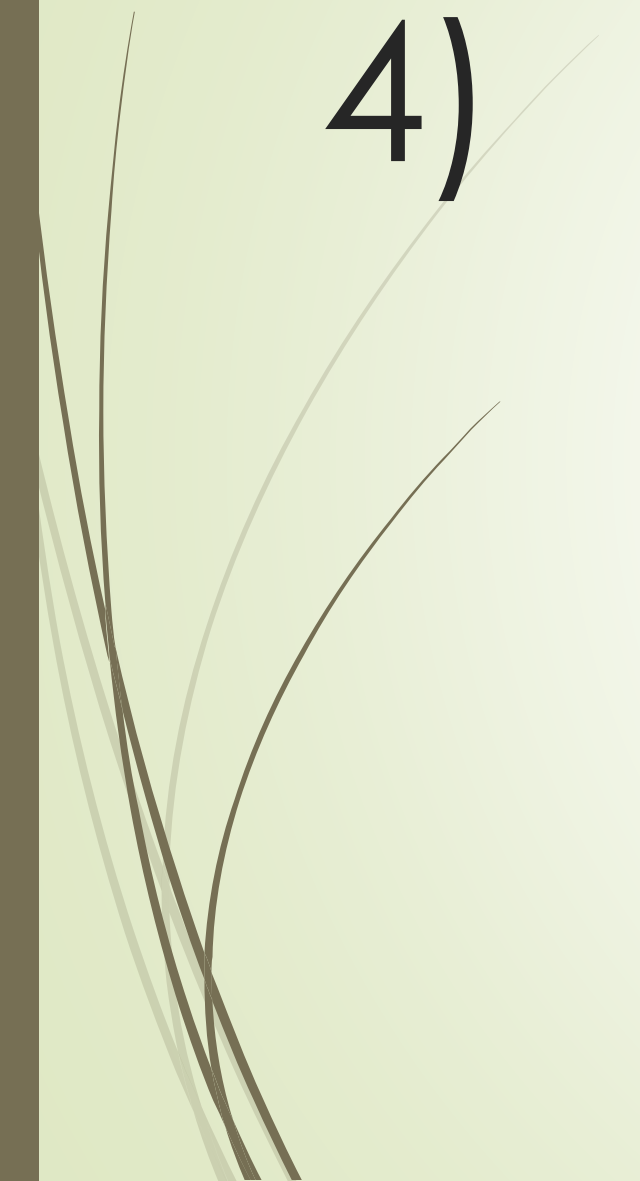
In computer security, Capture the Flag (CTF) is an exercise in which participants attempt to find text strings, called "flags", which are secretly hidden in purposefully vulnerable programs or websites.

- Pentest tools and media: [Pentest Tools & Media | Hak5 Official Site](#)

A penetration test (pentest) is an authorized simulated cyberattack on a computer system, performed live to evaluate the security of the system.



4)



References and Licenses

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